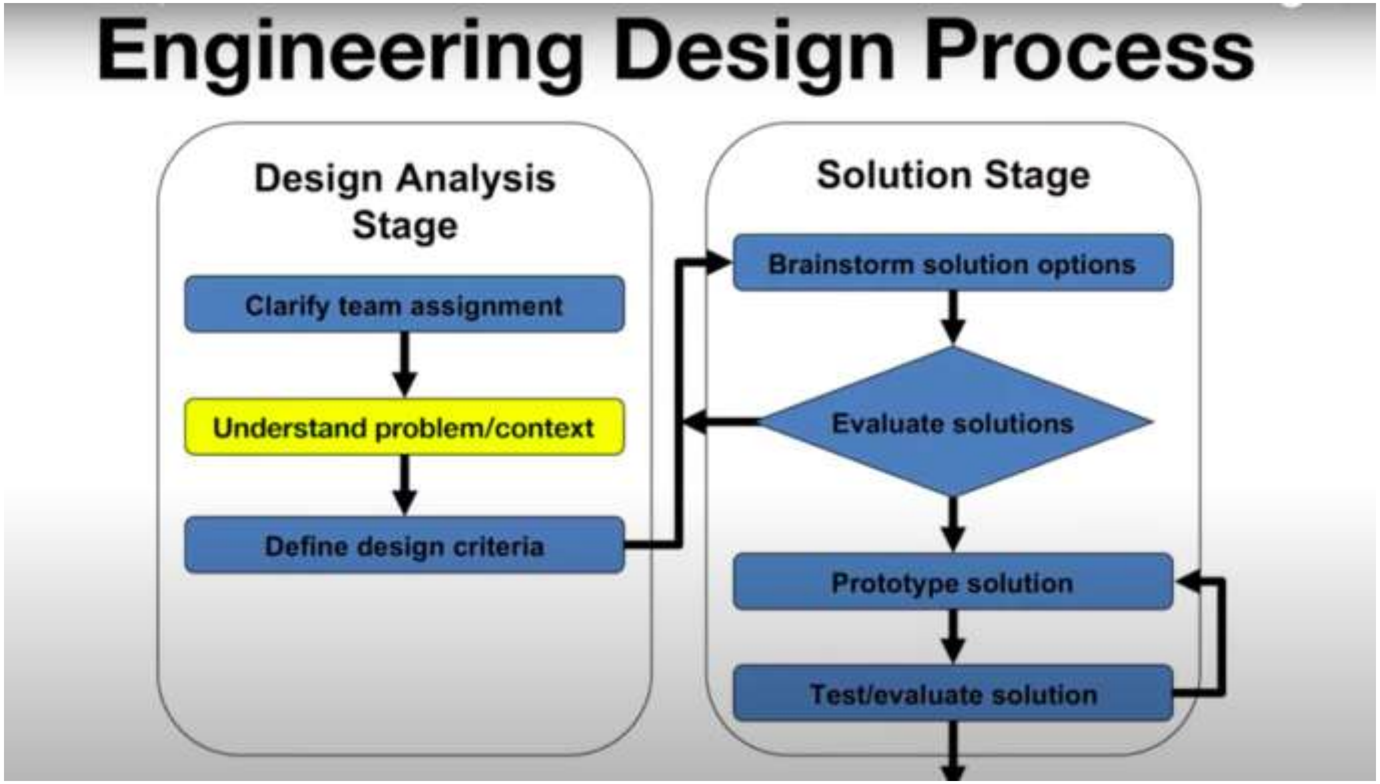
Understanding and Conceptualizing Interaction

Understanding The Problem Space



A Framework for Analysing the Problem Space

- Are there problems with an existing product?
- Why do you think there are problems?
- Why do you think your proposed ideas might be useful?
- How would you see people using it with their current way of doing things?
- How will it support people in their activities?
- Will it really help them?

An Example

- What were the assumptions made by cell phone companies when developing WAP services?

- Was it a solution looking for a problem?

Assumptions: realistic or wish-list?

- People want to be kept informed of up-to-date news wherever they are - reasonable
- People want to interact with information on the move - reasonable
- People are happy using a very small display and using an extremely restricted interface - not reasonable
- People will be happy doing things on a cell phone that they normally do on their PCs (e.g. surf the web, read email, shop, bet, play video games) - reasonable only for a very select bunch of users

From problem space to design space

- Having a good understanding of the problem space can help inform the design space
  - what kind of interface, behavior, functionality to provide
- But before deciding upon these, it is important to develop a conceptual model

Conceptual model

- A conceptual model is a high level description of:
  - "the proposed system in terms of a set of **integrated ideas and concepts about** <u>what it should do, behave and look like,</u> that will be understandable by the users in the manner intended"

First Steps in formulating a Conceptual Model

- What will the users be doing when carrying out their tasks?
- How will the system support these?
- What kind of interface metaphor, if any, will be appropriate?
- What kinds of interaction modes and styles to use?

Always keep in mind when making design decisions how the user will understand the underlying conceptual model .

Types of Conceptual Models (Ways Users Interact)

- Giving instructions
- Conversing
- Manipulating and navigating
- Exploring and browsing

1. Giving instructions

- Issuing commands using keyboard and function keys and selecting options via menus. (Usually through commands, buttons, or menus)
- Where users instruct the system and tell it what to do. (like tell the time, print a file, save a file)
- Main benefit is that instructing supports quick and efficient interaction. (good for repetitive kinds of actions performed on multiple objects)

2. Conversing

- Interacting with the system as if having a conversation
- Examples include timetables, search engines, advice-giving systems, help systems, AI generation response.
- Having virtual agents like chatbots and voice assistants.

3. Manipulating And Navigating

- Involves dragging, selecting, opening, closing and zooming actions on virtual objects
(Dragging, clicking, scrolling)
- Exploit's users' knowledge of how they move and manipulate in the physical world
- Shneiderman (1983) coined the term Direct Manipulation (DM).

3. Manipulating And Navigating

Core principles of Direct Manipulation(DM)

- Continuous representation of objects and actions of interest (Files, icons, buttons)
- Physical actions instead of typing(click, drag, swipe)
- Fast and reversible actions with feedback

4. Exploring And Browsing

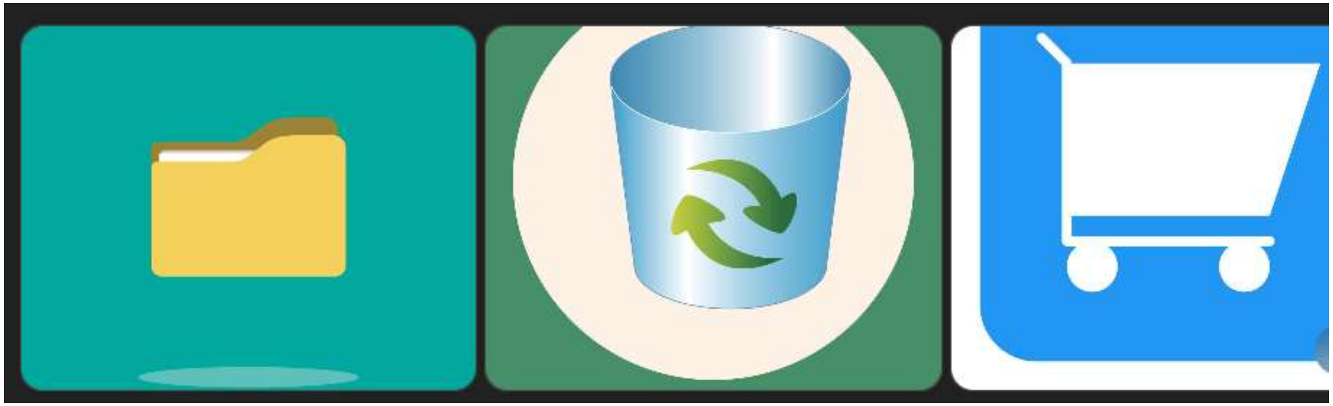
- Finding out and learning things
- Similar to how people browse information with existing media (newspapers, magazines, libraries)
- Information is structured to allow flexibility in way user is able to search for information.

Interface metaphors

Interface metaphor is when a system is designed **to look or behave like something familiar in real life**

Interface metaphors

- Interface **designed to be similar to a physical entity but also has own properties**
- Exploit user’s familiar knowledge, helping them to understand ‘the unfamiliar’
- Use familiar ideas to understand unfamiliar technology



Benefits of Interface Metaphors

- Makes learning new systems easier
- Helps users understand the underlying conceptual model
- Can be very innovative and enable the realm of computers and their applications to be made more accessible to a greater diversity of users.

Problems With Interface Metaphors

- Can constrain designers in the way they conceptualize a problem space
- Limits designers
- Forces users to only understand the system in terms of the metaphor
- Limits designers’ imagination in coming up with new conceptual models

Conceptual Models: From Interaction Mode to Style

- Interaction mode:
  - *what the user is doing* when interacting with a system, like instructing, talking, browsing or other
- Interaction style:
  - the kind of *interface used to support* the mode, like speech, menu-based, gesture

Many kinds of interaction styles available…

- Command
- Speech
- Data-entry
- Form fill-in
- Query
- Graphical
- Web
- Pen
- Augmented reality
- Gesture

Which interaction style to choose?

1. Need to determine requirements and user needs
2. Take the budget and other constraints into account
3. Also, will depend on suitability of technology for activity being supported

Interaction paradigms

It is the **different ways people interact with computers or systems.**

Like **styles or approaches in designing** how users use technology.

Examples of New Paradigms

- Pervasive computing
- Wearable computing
- Tangible bits, augmented reality
- Attentive environments
- Transparent computing

Summary points

- Important to have a good understanding of the problem space
- Fundamental aspect of interaction design is to develop a conceptual model
- Interaction modes and interface metaphors provide a structure for thinking about which kind of conceptual model to develop
- Interaction styles are specific kinds of interfaces that are instantiated as part of the conceptual model
- Interaction paradigms can also be used to inform the design of the conceptual model